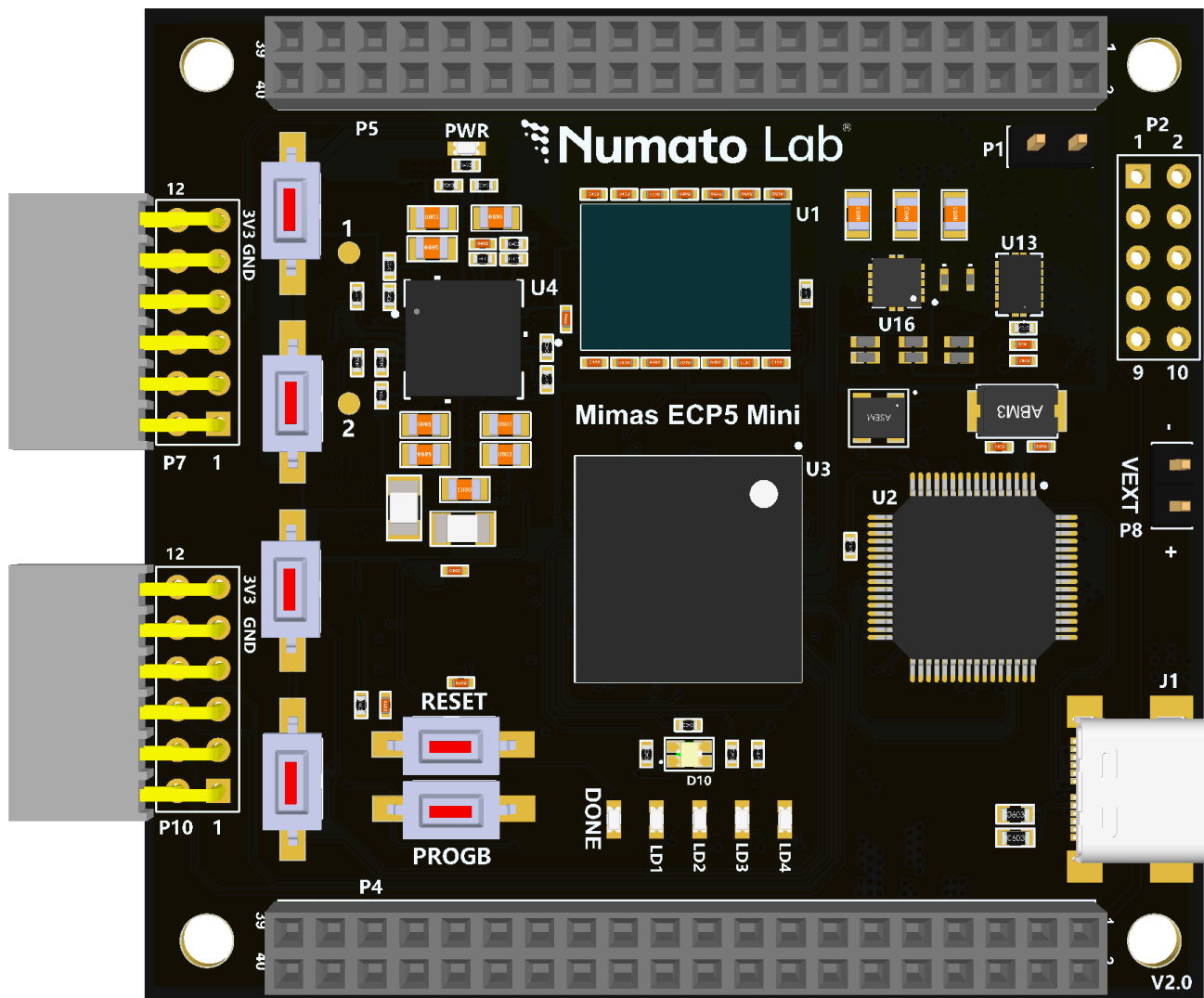


Introduction



(https://numato.com/help/wp-content/uploads/2022/01/Mimas_ECP5_mini.png)

Mimas ECP5 Mini FPGA Development board is a simple FPGA board having Lattice ECP5 FPGA (LFE5U-45F-7BG256I package) with FTDI's FT2232H Dual-Channel USB device. The Lattice ECP5 FPGA family is renowned for breaking the rules of power, size, and cost in your connectivity and acceleration applications. The USB 2.0 host interface based on the popular FT2232H offers high bandwidth data transfer and board programming without the need for any external programming adapters.

Board Features

- Device: Lattice ECP5 FPGA (LFE5U-45F-7BG256I)
- DDR3: 2Gb DDR3 (MT41J128M16JT-125 or equivalent)
- Built-in programming interface. No expensive JTAG adapters are needed for programming the board
- Onboard 128Mb flash memory for FPGA configuration storage and custom user data storage

- High-speed USB 2.0 interface for Onboard flash programming.
- 4 LEDs, 1 RGB LED, and 4 Push Buttons for user-defined purposes
- FPGA configuration via JTAG and USB
- Maximum IOs for user-defined purposes
 - FPGA – 70 IOs (35 professionally length matched Differential Pairs) and two 2×6 Expansion Headers

Applications

- Product Prototype Development
- Accelerated computing integration
- Development and testing of custom-embedded processors
- Communication devices development
- Educational tool for Schools and Universities

How to use Mimas ECP5 mini FPGA development Board

The following section describe in detail how to use this module

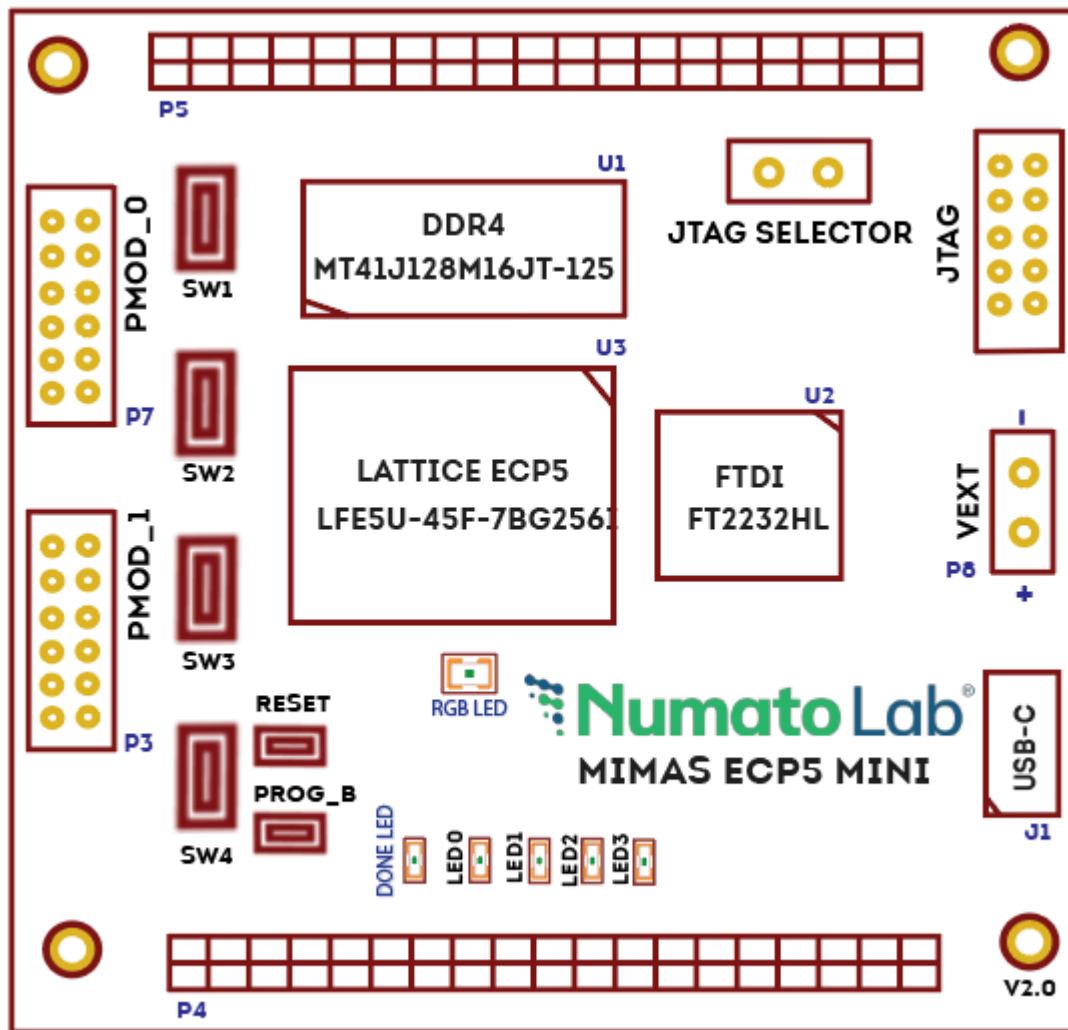
Hardware Accessories Required

For easy and fast installation, you may need the following items along with the Mimas ECP5 Mini module.

- USB B Micro cable
- DC Power supply

Connection Diagram

The following connection diagram should be used for reference only. The schematics are available at the end of this document for detailed information.



(<https://numato.com/help/wp-content/uploads/2022/01/ecp5.png>)

USB Interface

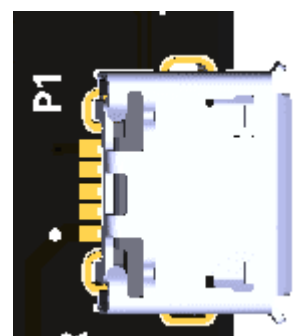
The onboard full-speed USB controller helps a PC/Linux/Mac computer to communicate with this module. Use a USB A to USB B Micro cable to connect with a PC (the picture given below shows the USB B Micro connector).

By default, the module is powered by USB so make sure not to overcrowd unpowered USB hubs.

Note: Mimas S7 lite ships with FT2232H Channel A dedicated to JTAG Programming.

External Power Supply

The board can be configured to use an External power supply by connecting it to the External +5V supply. Please refer to the marking on the board for more details (the picture on the right shows the External +5V supply connector).



PROG_B and Reset Buttons

Mimas ECP5 Mini features a Push-button **PROG_B** normally used as a “PROG_B” signal for configuration reset. Push-button **PROG_B** is connected to FPGA pin **R9**. For enabling manual configuration reset, push-button **PROG_B** is connected to GND. The user can reconfigure the FPGA manually, by pressing this push-button **PROG_B**.

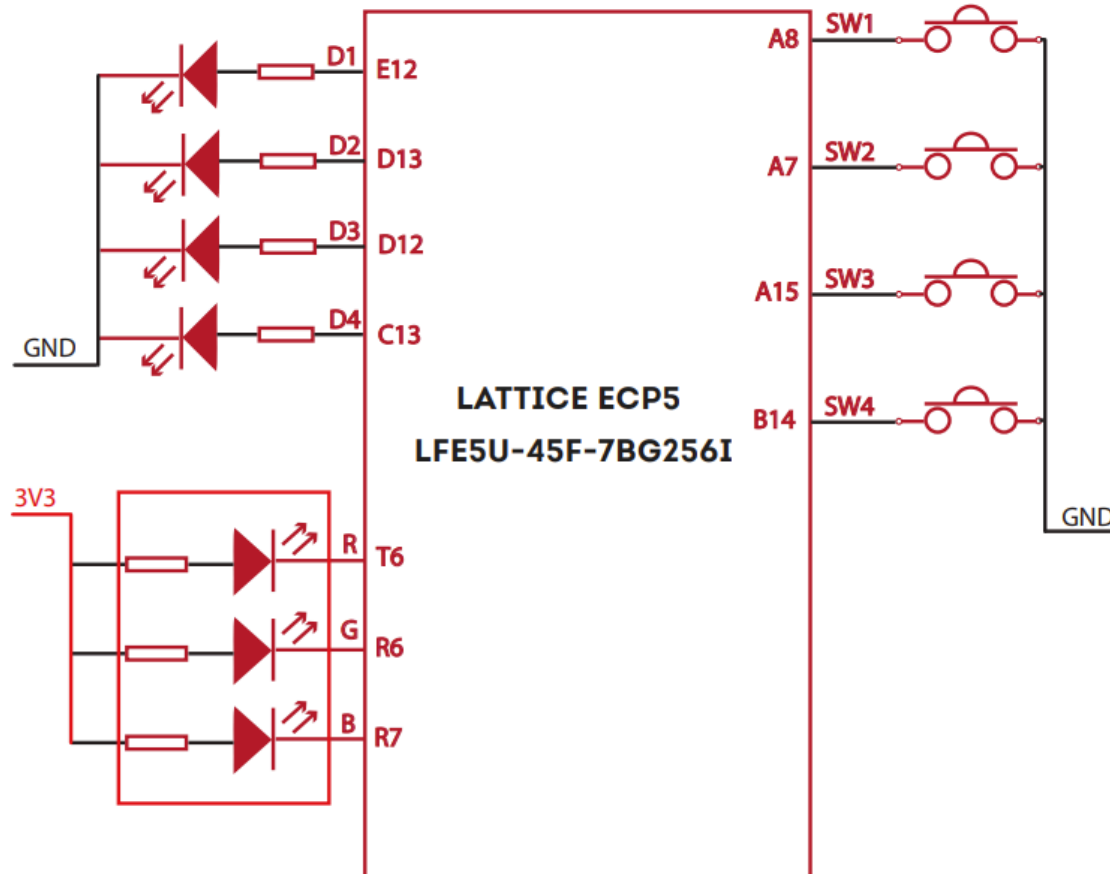
“PROG_B” controls the configuration logic. When the PROG_B pin is de-asserted, it resets the FPGA and initializes the new configuration.

Mimas ECP5 Mini features a Push-button **RESET** normally meant to be used as a “Reset” signal for designs running on FPGA. Push-button **RESET** is connected to FPGA pin **A4**. Push-button **RESET** is **active-high**. This push button can also be used for any other input and is not limited to being used as a Reset signal.



LEDs, RGB LED and Push Button

Mimas ECP5 Mini Development Board has four push-button switches, one RGB LED, and four LEDs for human interaction. All switches are directly connected to Lattice ECP5 FPGA and can be used in your design with minimal effort.



(<https://numato.com/help/wp-content/uploads/2022/01/Screenshot-2024-08-26-105032.png>)

GPIOs

This device is equipped with a maximum of 70 user IO pins that can be used for various custom applications. All user IOs are length matched and can be used as differential pairs.

Header P4

Pin No. On the Header	GPIO Pin Name	ECP5 (7BG256I) Pin No.	Pin No. On the Header	GPIO Pin Name	ECP5 (7BG256I) Pin No.
2		GND	1		VCCIO
4	GPIO_1_P	B16	3	GPIO_1_N	B15
6	GPIO_2_P	C16	5	GPIO_2_N	C15
8	GPIO_3_P	D16	7	GPIO_3_N	E15
10	GPIO_4_P	G12	9	GPIO_4_N	G13
12	GPIO_5_P	F16	11	GPIO_5_N	G15
14	GPIO_6_P	G16	13	GPIO_6_N	H15
16	GPIO_7_P	H12	15	GPIO_7_N	H13
18	GPIO_8_P	K13	17	GPIO_8_N	K12
20	GPIO_9_P	J16	19	GPIO_9_N	J15
22	GPIO_10_P	J14	21	GPIO_10_N	K14
24	GPIO_11_P	K16	23	GPIO_11_N	K15
26	GPIO_12_P	L16	25	GPIO_12_N	L15
28	GPIO_13_P	N16	27	GPIO_13_N	P15
30	GPIO_14_P	P16	29	GPIO_14_N	R16
32	GPIO_15_P	N13	31	GPIO_15_N	P14
34	GPIO_16_P	P13	33	GPIO_16_N	R14
36	GPIO_18	R12	35	GPIO_17	T13
38	GPIO_20	N11	37	GPIO_19	M11
40		GND	39		VCCIO

Header P5

Pin No. On the Header	GPIO Pin Name	ECP5 (7BG256I) Pin No.	Pin No. On the Header	GPIO Pin Name	ECP5 (7BG256I) Pin No.
2		VCCIO			GND
4	GPIO_22	E16	3	GPIO_21	C14
6	GPIO_24	F15	5	GPIO_23	D14
8	GPIO_26	E14	7	GPIO_25	F14
10	GPIO_28	F12	9	GPIO_27	F13
12	GPIO_30	C9	11	GPIO_29	B9
14	GPIO_32	E9	13	GPIO_31	D9
16	GPIO_34	C8	15	GPIO_33	B8
18	GPIO_36	E8	17	GPIO_35	D8
20	GPIO_38	J13	19	GPIO_37	J12
22		GND	21		GND
24	GPIO_40	L13	23	GPIO_39	L12
26	GPIO_42	M12	25	GPIO_41	N12
28	GPIO_44	P12	27	GPIO_43	P11
30	GPIO_46	M14	29	GPIO_45	M13
32	GPIO_48	R13	31	GPIO_47	N14
34	GPIO_50	L14	33	GPIO_49	M15
36	GPIO_52	T15	35	GPIO_51	T14
38	GPIO_54	M16	37	GPIO_53	R15
40		VCCIO	39		GND

Header P10 (2x6 Expansion Header)

Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	ECP5 (7BG256I) Pin No.	Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	ECP5 (7BG256I) Pin No.
6		VCC	12		VCC

Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	ECP5 (7BG256I) Pin No.	Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	ECP5 (7BG256I) Pin No.
5		GND	11		GND
4	CONN1_D3	C10	10	CONN1_D7	A14
3	CONN1_D2	B10	9	CONN1_D6	A13
2	CONN1_D1	E10	8	CONN1_D5	A12
1	CONN1_D0	D10	7	CONN1_D4	A11

Header P7 (2x6 Expansion Header)

Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	ECP5 (7BG256I) Pin No.	Pin No. On the Header	GPIO 2x6 CONNECTOR Pin Name	GPIO 2x6 CONNECTOR Pin Name
6		VCC	12		VCC
5		GND	11		GND
4	CONN0_D3	A10	10	CONN0_D7	G14
3	CONN0_D2	A9	9	CONN0_D6	H14
2	CONN0_D1	E11	8	CONN0_D5	C12
1	CONN0_D0	D11	7	CONN0_D4	B12

FT2232H – ECP5(7BG256I) FPGA Connection Detail

FTDI Pin No.	Pin Function (245 FIFO)	ECP5 (7BG256I) Pin No.
38	FTDI_D0	E6
39	FTDI_D1	D6
40	FTDI_D2	A5
41	FTDI_D3	A6
43	FTDI_D4	C5
44	FTDI_D5	B5
45	FTDI_D6	E5
46	FTDI_D7	D5

FTDI Pin No.	Pin Function (245 FIFO)	ECP5 (7BG256I) Pin No.
48	FTDI_RXE_N	C4
52	FTDI_TXE_N	B4
53	FTDI_RD_N	E4
54	FTDI_WR_N	D4
55	FTDI_SIWUA_N	A3

Driver Installation

Windows

This product requires a D2XX driver to be installed for proper functioning when used with Windows. The driver can be downloaded from D2XX drivers (<https://ftdichip.com/drivers/d2xx-drivers/>). When the driver installation is complete, the module should appear in FT_Prog Tool as Mimas ECP5 Mini FPGA Development Board.

Linux

The Linux ships with the drivers required for Mimas ECP5 Mini.

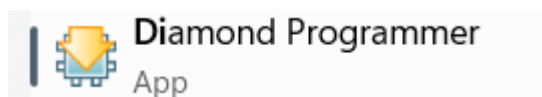
Programming Mimas ECP5 Mini

Step 1:

Before Programming make sure to **populate** Jumper **P1**. Power up Mimas ECP5 Mini Development board by connecting the USB C cable.

Step 2:

Open “Diamond Programmer.

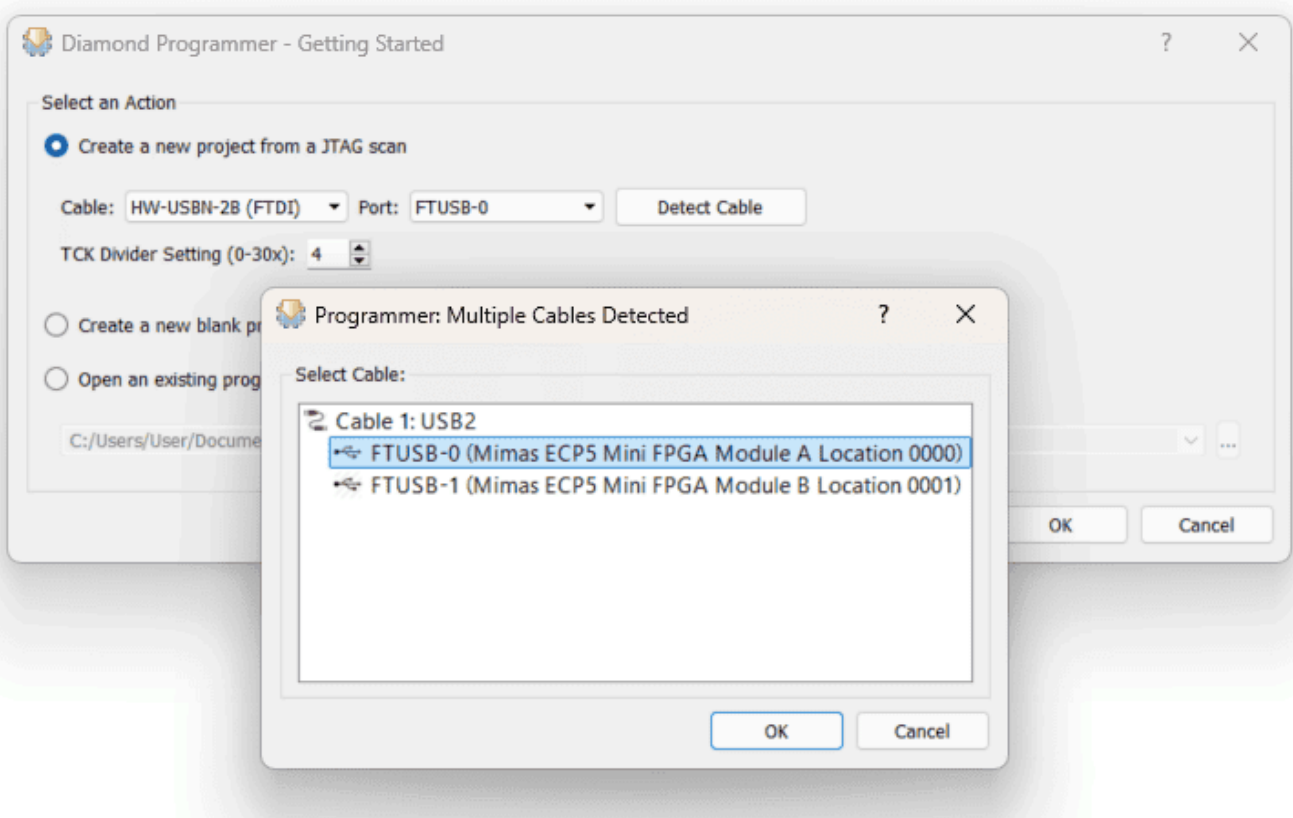


([https://numato.com/help/wp-](https://numato.com/help/wp-content/uploads/2024/08/Screenshot-2024-04-02-110617.png)

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Step 3:

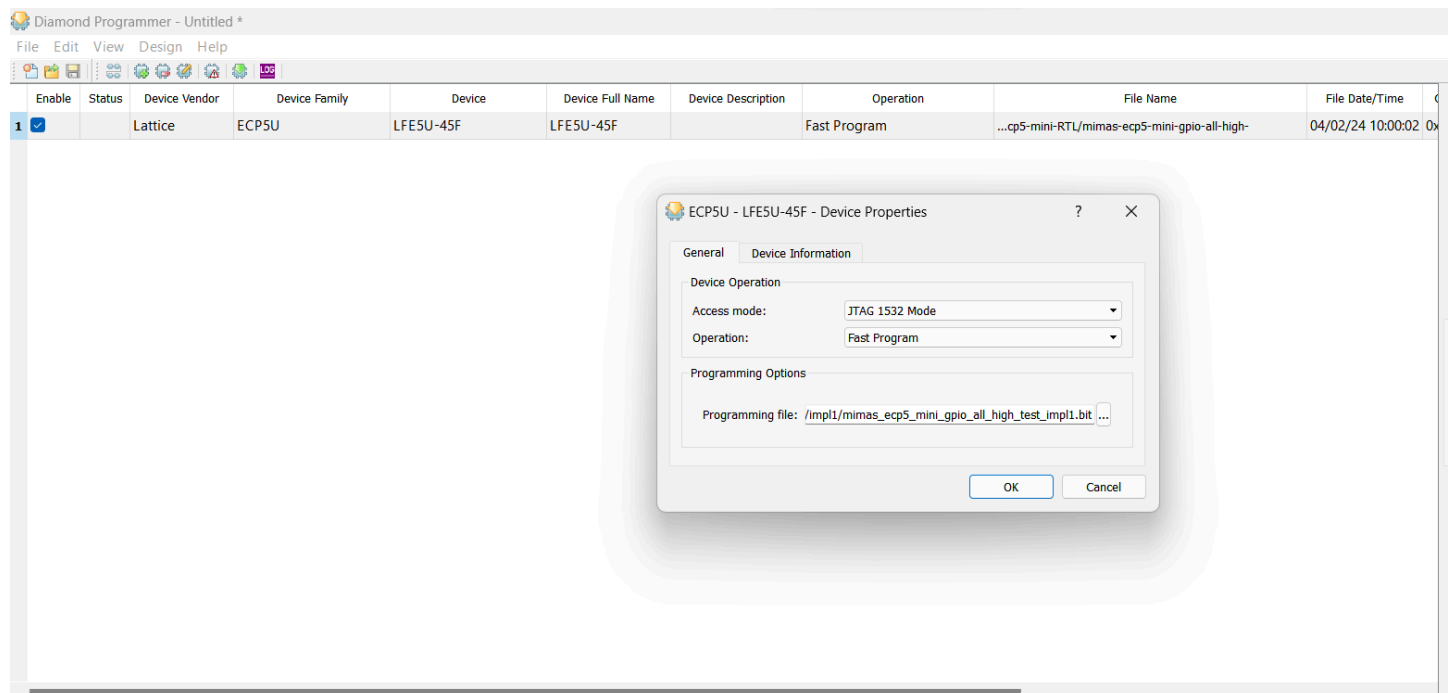
Click on “Detect Cable” and select the correct port. Click “OK”



(<https://numato.com/help/wp-content/uploads/2024/08/Screenshot-2024-08-28-133025.png>)

Step 4:

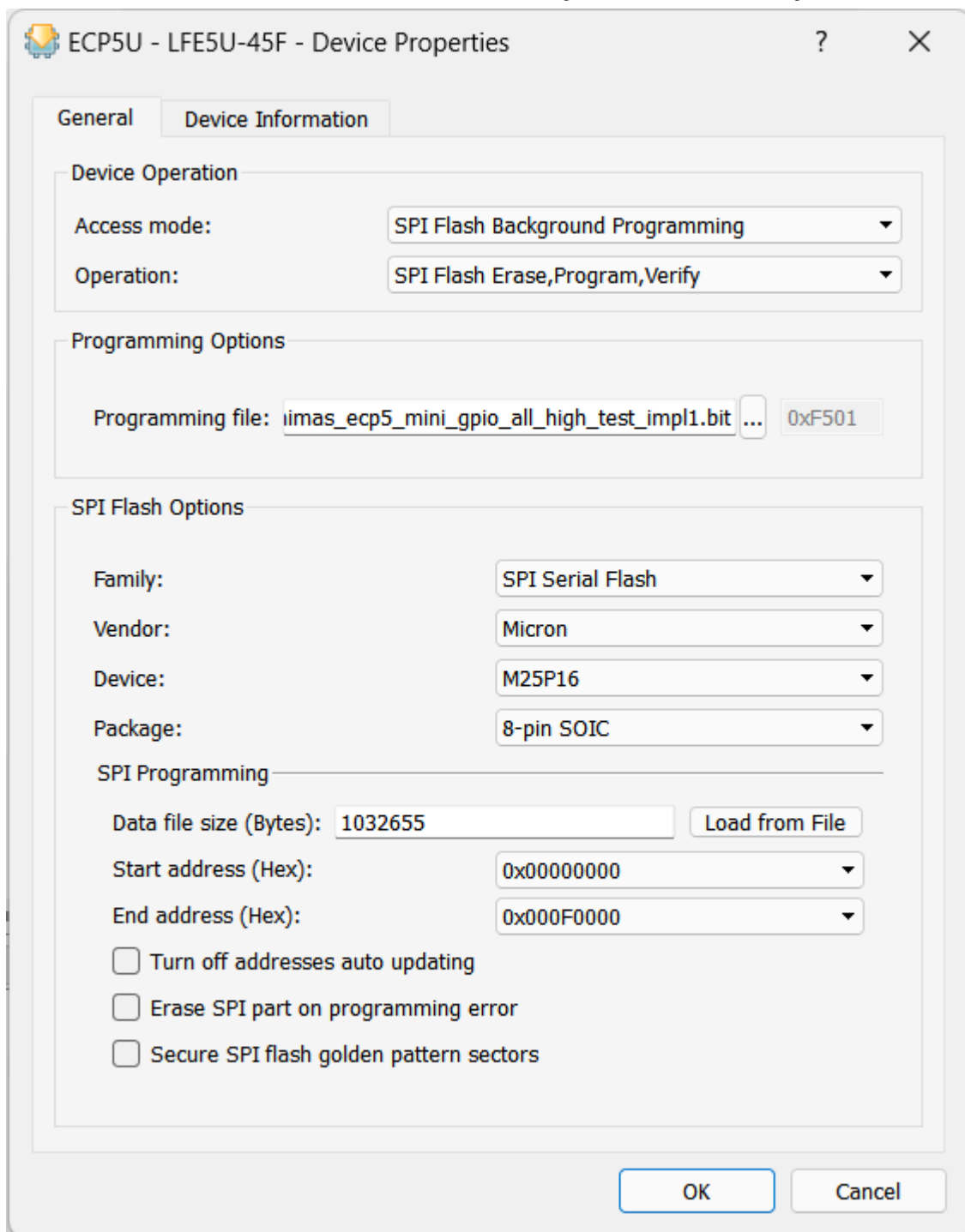
Select the Operation as “Fast Program” and browse the Corresponding Bitstream file in the File Name box.



(<https://numato.com/help/wp-content/uploads/2024/08/Screenshot-2024-04-02-112238.png>)

Step 5:

For programming SPI Flash Select the “Operation” as given below.



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Step 6:

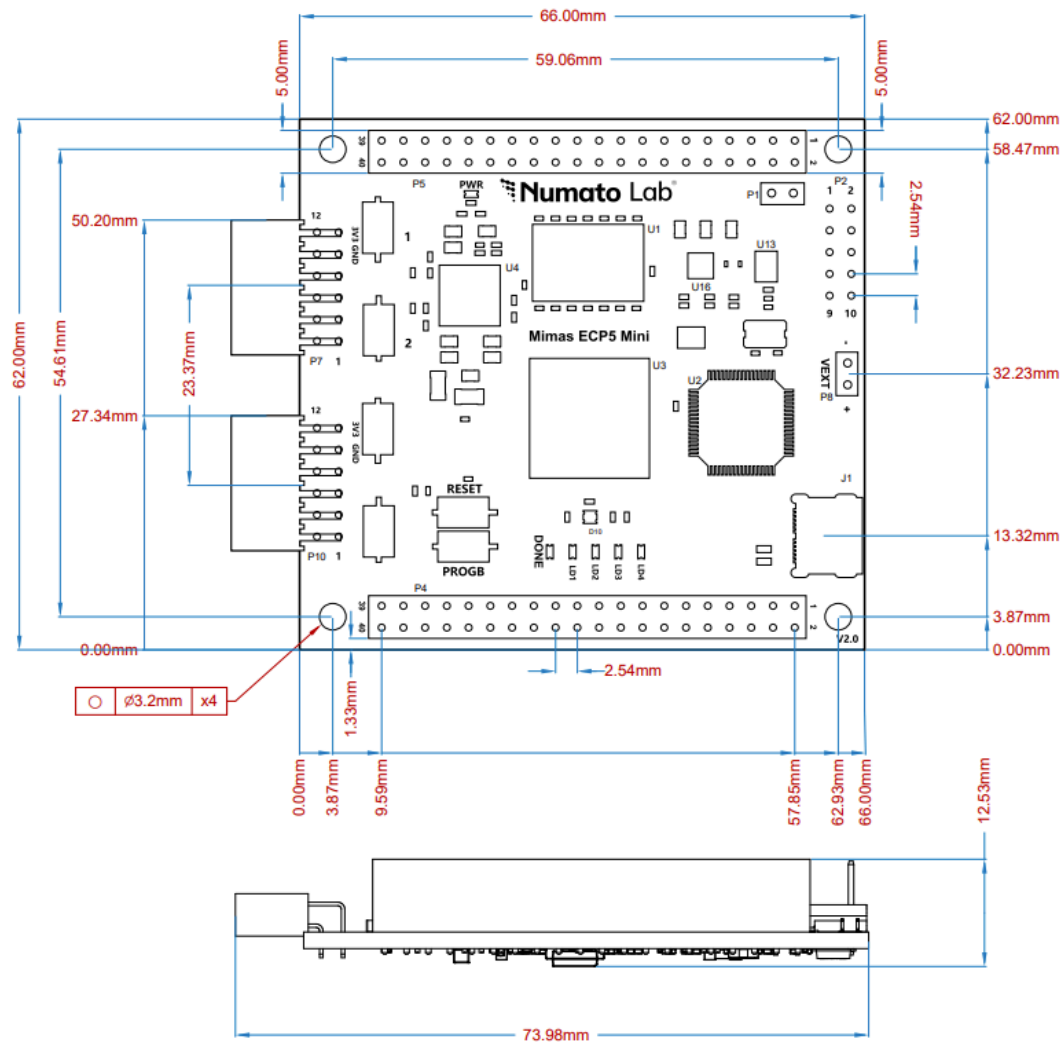
Finally, click on “Program” ().

Note: SPI Flash Background Programming is only possible when the ECP5-5G device is erased or when the active design has the MASTER_SPI_PORT mode enabled. Therefore, when performing SPI Flash Background Programming, ensure the PROGB button is held down until programming is completed.

Technical Specifications

Parameter *	Value	Unit
Basic Specifications		
Number of GPIOs	70	
On-board oscillator frequency (ASEM1-100.000MHZ-LC-T)	100	MHz
DDR3 SDRAM (MT41J128M16JT-125 or equivalent)	2	Gb
Quad SPI Flash Memory (M25P16)	128	Mb
Power supply voltage (USB or External)	5	V
Number of LEDs	4	
Number of Push Buttons	4	
FPGA Specifications		
Internal supply voltage relative to GND	-0.5 to 1.1	V
Auxiliary supply voltage relative to GND	-0.5 to 2.0	V
Output drivers supply voltage relative to GND	-0.5 to 3.6	V
Temperature range	0 to 70	°C

Physical Dimensions



(<https://numato.com/help/wp-content/uploads/2022/02/Screenshot-2024-08-26-105223.png>)

Schematics

Mimas_ECP5_Mini_Schematics (https://numato.com/help/wp-content/uploads/2022/02/Mimas_ECP5_Mini_SCH_V2.1.pdf)

Mimas ECP5 Mini GPIO Easy Reference

Mimas_ECP5_mini_GPIO_easy_reference (https://numato.com/help/wp-content/uploads/2022/02/Mimas_ECP5_mini_GPIO_easy_reference.pdf)

Mimas ECP5 Mini IO length Details

Mimas ECP5 Mini IO length details (<https://numato.com/help/wp-content/uploads/2024/08/IO-length.pdf>)

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